

LABORATORY REPORT - CHAPTER 4

v7.4

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Total Grade	/100

Record all your measurements and write all your answers in the boxes provided.

Preliminary Work

1. First IF Amplifier

1. An RF amplifier can be built using a BJT. Refer to the circuit given in Fig. 1. TRC-11 uses two of such amplifiers. The values of the resistors and capacitors should be calculated as described below.

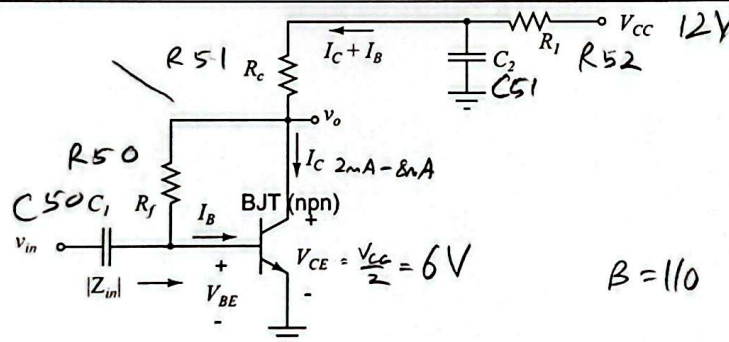


Figure 1: Schematic of an RF amplifier using a BJT.

2. BJT to be used is KSP10, an NPN transistor. Take a look at its datasheet given in 344. It is a high-frequency transistor. β (or h_{FE}) of the transistor is a minimum of 60. Once the operating point (collector current, I_C , and the collector-emitter voltage, V_{CE}) is set, the values of the bias resistors can be calculated as follows:

$$R_{51} + R_{52} = \frac{V_{CC} - V_{CE}}{I_C + I_B} \approx \frac{V_{CC} - V_{CE}}{I_C} = \frac{6V}{I_C} = 1.2k\Omega / (1) 800\Omega$$

$$R_{50} = \frac{V_{CE} - V_{BE}}{I_B} \approx \beta \frac{V_{CE} - 0.8}{I_C} = \frac{5.2V}{I_C} = 114.4k\Omega / (2) 76.2k\Omega$$

where we assumed $V_{BE} \approx 0.8$. For the first amplifier, calculate the values of resistors, $R_{50} = R_f$ and $R_{51} + R_{52} = R_c + R_1$, choosing a DC operating point of I_C a value in the

range 2 mA to 8 mA and $V_{CE}=V_{CC}/2$. The operating voltage V_{CE} is chosen to be half the supply voltage to provide a maximum voltage swing at the collector. Choose $V_{CC}=12$ V. Take $\beta=110$ for the purpose of this calculation. Choose $R_{52} = R_1$ in the range 22 to 56 Ω . Then find the value of R_c from Eq. 1. Similarly, determine the value of R_f from Eq. 2. Select the resistors by rounding the numbers to the closest standard resistor values: 1, 1.2, 1.5, 1.8, 2.2, 2.7, 3.3, 3.9, 4.7, 5.6, 6.8, and 8.2.

3. The DC block capacitor, C_1 , at the input is used to block DC voltages from affecting the base voltage of the transistor. Its value should be large enough so that there is no AC voltage drop across it. Its value can be found from

$$C_1 > \frac{10}{2\pi f_0 |Z_{in}|} \quad (3)$$

where f_0 is the operating frequency of 15 MHz. Assume $|Z_{in}|=50 \Omega$ and find $C_{50}=C_1$ from Eq. 3, with all variables in MKS units. Round its value to the standard capacitor values: 1, 1.2, 1.5, 1.8, 2.2, 2.7, 3.3, 3.9, 4.7, 5.6, 6.8, and 8.2.

4. The power supply bypass capacitor, C_2 , and the resistor, R_1 , forms a low-pass-filter. They are used to provide a clean power supply voltage to the amplifier. Choose a corner frequency f_1 at least 100 times smaller than f_0 .

$$C_2 \approx \frac{1}{2\pi f_1 R_1} \quad (4)$$

Choose $C_{51}=C_2$ using the standard capacitor values.

5. Repeat the procedure for the second IF amplifier. For the second amplifier, choose I_C in the range 5 mA to 10 mA (higher than the first).

First amplifier: $I_C = 5 \text{ mA}$	$V_{CE} = 6 \text{ V}$
$R_{50} = 120 \text{ k}\Omega$	$R_{51} = 1.2 \text{ k}\Omega$ $R_{52} = 33 \Omega$
$C_{50} = 10 \text{ nF}$	$C_{51} = 330 \text{ nF}$
Second amplifier: $I_C = 7.5 \text{ mA}$	$V_{CE} = 6 \text{ V}$
$R_{60} = 82 \text{ k}\Omega$	$R_{61} = 820 \Omega$ $R_{62} = 33 \Omega$
$C_{60} = 10 \text{ nF}$	$C_{61} = 330 \text{ nF}$

1.5. GRADE:

6. The gain of the BJT amplifiers can be calculated using LTSpice. SPICE model of the KSP10 is given as

```
.model KSP10TA NPN(IS=1E-11 ISE=70e-11 NE=2.5 VAF=100 BF=110
+ NF=1.3 IKF=0.065 NK=0.45 XTB=0 BR=3 CJC=1.56E-12
+ CJE=2E-12 TR=50e-9 TF=1.85E-10 ITF=0 VTF=0 XTF=0
+ RB=42 RC=.3 RE=.2 Vceo=25 Icrating=100m mfg=0nSemi)
```


Place these lines in LTSpice as a SPICE directive. Define the NPN transistor's "Value" (seen after CTRL-right click on the transistor) as KSP10TA (the same label used in the model statement). Enter the schematic of the first amplifier in LTSpice using the calculated values of components. Determine the DC voltages V_{CE} and V_{BE} using "DC opt pnt" analysis.

7. Determine $|Z_{in1}|$ of the first amplifier at 15 MHz by connecting an AC current source of unity magnitude and performing a small-signal AC analysis. The voltage of the current source gives the input impedance, $|Z_{in}|$. Change the default display of the y-axis to "Bode/linear" to read the magnitude of the impedance (specified in V). Refer to Appendix on LTSpice Tutorial on page 399.
8. Determine the small-signal gain, $|v_{o1}/v_{in1}|$, in dB at 15 MHz also by using small-signal AC analysis. You should define an AC voltage source of unity magnitude at the input with a series source resistance of $50\ \Omega$. (The signal generators at the lab has an internal source resistance of $50\ \Omega$.) Place a 20 pF capacitor to ground at the output to simulate the probe capacitance of the oscilloscope (to be compared with measurements later). Set the y-axis to its default: "Bode/decibel" to read the gain in dB.

$$V_{CE1} = 7.21\text{ V} \quad V_{BE1} = 0.67\text{ V}$$

$$|Z_{in1}| = 108.65\ \Omega \quad |v_{o1}/v_{in1}|_{dB} = 30.7\text{ dB}$$

1.8. GRADE:

2. The second IF Amplifier

1. Repeat the LTSpice simulations for the second amplifier.

$$V_{CE2} = 7.05\text{ V} \quad V_{BE2} = 0.68\text{ V}$$

$$|Z_{in2}| = 101.75\ \Omega \quad |v_{o2}/v_{in2}|_{dB} = 32.20\text{ dB}$$

2.1. GRADE:

2. Connect the second amplifier's input to the first amplifier's output. Determine the input impedance, the gain of the first amplifier and the overall gain in dB at 15 MHz. Note that the input impedance and the gain of the first amplifier is not the same as that determined when the amplifier is stand-alone. The reason is that the output load impedance of the first amplifier is changed. The input of the second amplifier is loading the output of the first amplifier, reducing the gain of the first amplifier and also changing the input impedance. The input impedance of a BJT is influenced by the load impedance at the output, especially at high frequencies. At very low frequencies, the input impedance does not get affected by the output load.

$$|Z_{in1}| = 299.05\ \Omega \quad |v_{o1}/v_{in1}|_{dB} = 21.99\text{ dB}$$

$$\text{Overall gain: } |v_{o2}/v_{in1}|_{dB} = 57.32\text{ dB}$$

3. 2.3. GRADE:

Experimental Work

1. IF Amplifiers

1. The first RF amplifier circuit to be built is the first IF amplifier.

A schematic diagram of the first IF amplifier is given in Fig. 1. Mount the three leads of

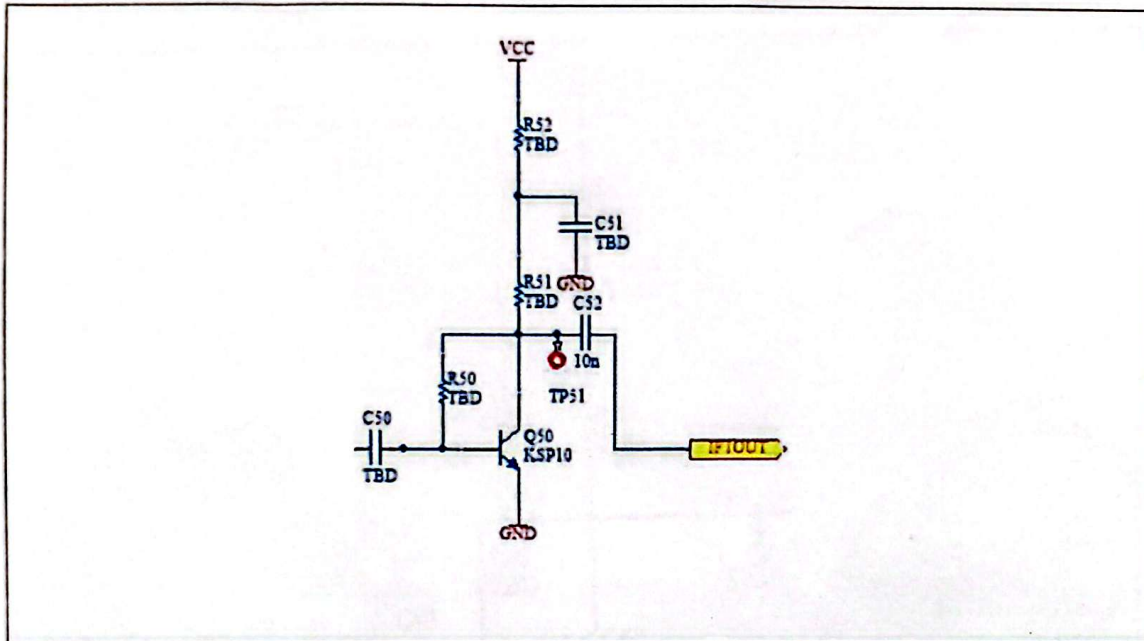


Figure 2: Schematic of the first IF amplifier.

Designator	Comment	Description
C52	10 nF	Capacitor, ceramic disc, 50V
C53	10 nF	Capacitor, ceramic disc, 50V
Q50	KSP10	NPN Bipolar transistor

Figure 3: Bill of materials for the first IF amplifier

the transistor Q50 (KSP10) watching the orientation on the PCB (e: Emitter, b: Base, c: Collector) and solder it. If you need to look at the datasheet, use only the datasheet in the textbook. Some datasheets that can be found in internet shows the emitter and collector leads swapped, resulting in non-functioning circuit.

Do not push the transistor too hard into the holes. Clip the extra leads at the back side.

2. Mount and solder C53. This is a supply bypass capacitor.

3. Using the chosen values collect the resistors R50, R51, R52, and the capacitors, C50, C51. Measure the resistors and capacitors before you mount them. Note that the capacitors may have a tolerance up to 20%. This is not a problem. Mount the resistors and the capacitors. Solder them and clip the leads on the back side. Do not mount C52 yet.
4. Solder a clipped resistor lead between the GND terminals (at the right of C50) for a convenient GND connection later.
5. Solder bent resistor leads at the test points, TP50 and TP51.
6. Second IF amplifier is to be mounted next. A schematic diagram of the second IF amplifier is given in Fig. 6.

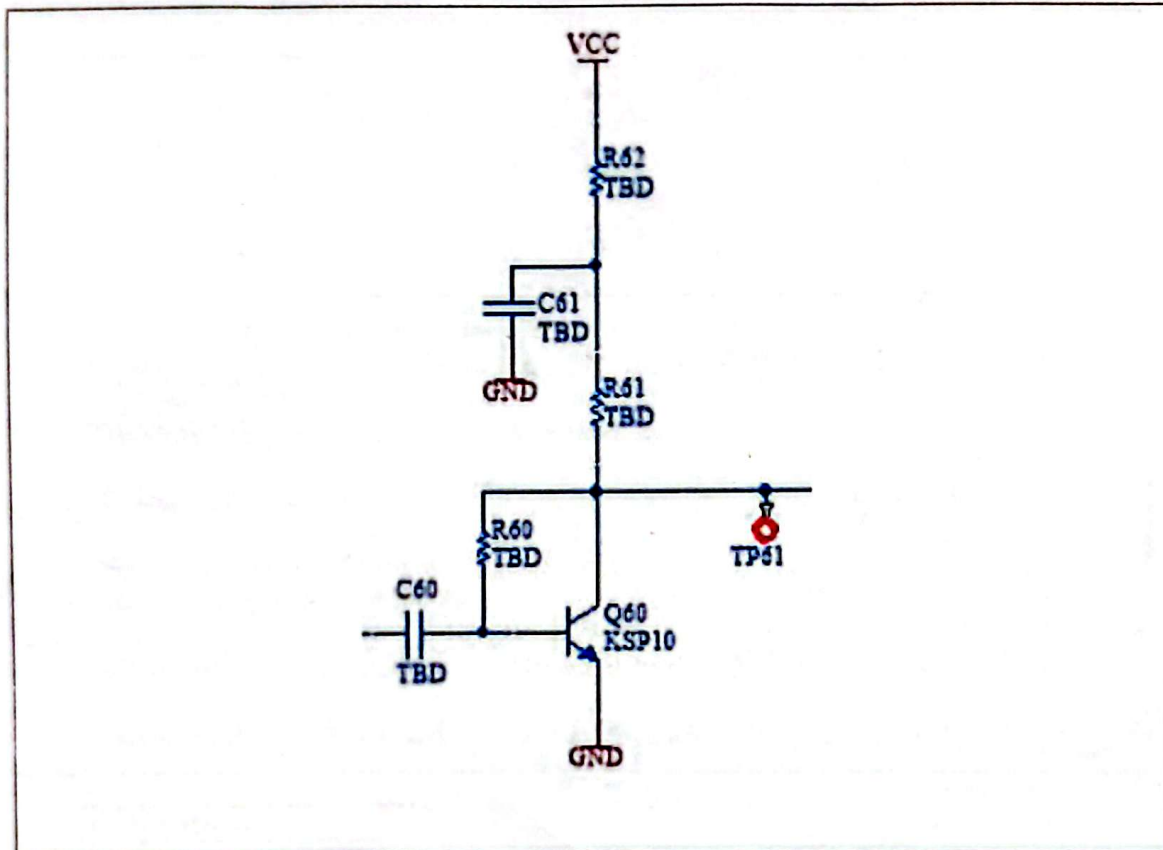


Figure 4: Schematic of the second IF amplifier.

Designator	Comment	Description
C66	10 nF	Capacitor, ceramic disc, 50V
Q60	KSP10	NPN Bipolar transistor

Figure 5: Bill of materials for the second IF amplifier

7. Mount the components of the second IF amplifier: Mount and solder C66. This is a supply bypass capacitor.

8. Mount and solder Q60 (KSP10) watching the orientation of the transistor on the PCB.
9. Collect the resistors R60, R61, R62, and the capacitors, C60 and C61. Measure the resistors and capacitors before you mount them. Solder all components.
10. Solder a clipped resistor lead between the GND terminals (near C65) for convenient connection later.
11. Solder bent resistor leads at the test points, TP60 and TP61.
12. Apply the supply voltage using the adapter. Measure the DC voltages, V_{CE} (between TP51 and GND for Q50; between TP61 and GND for Q60) at the collectors of BJTs for both amplifiers. Measure the base to emitter voltages, V_{BE} , for both transistors. Using the measured values of resistors, and voltage measurements calculate the actual β of the transistors:

$$I_C + I_B = \frac{V_{CC} - V_{CE}}{R_{51} + R_{52}} \quad (5)$$

$$I_B = \frac{V_{CE} - V_{BE}}{R_{50}} \quad (6)$$

$$\beta = \frac{I_C}{I_B} \quad (7)$$

First amplifier: Measured values of resistors

$$R_{50} = 117.812 \Omega \quad R_{51} = 1189 \Omega \quad R_{52} = 32.6 \Omega$$

$$\text{Measured } V_{BE1} = 0.724 \text{ V} \quad \text{Measured } V_{CE1} = 5.72 \text{ V}$$

$$\text{Calculated } I_{C1} = 5.141 \text{ mA} \quad I_{B1} = 41.986 \mu\text{A} \quad \beta_1 = 121.44 \text{ dB}$$

Second amplifier: Measured values of resistors

$$R_{60} = 79.94 \Omega \quad R_{61} = 803.1 \Omega \quad R_{62} = 33.5 \Omega$$

$$\text{Measured } V_{BE2} = 0.729 \text{ V} \quad \text{Measured } V_{CE2} = 5.72 \text{ V}$$

$$\text{Calculated } I_{C2} = 7.506 \text{ mA} \quad I_{B2} = 61.952 \mu\text{A} \quad \beta_2 = 120.17 \text{ dB}$$

1.12. GRADE:

13. Set the signal generator to 15 MHz, 50 mVpp sinusoidal signal between TP50 and GND (connect the black lead to GND). Most high frequency signal generators assume that the load resistance is equal to their source resistance of 50Ω , and hence their display shows half the internal voltage value. That means we are actually using a 100 mVpp voltage source in series with 50Ω resistance. Measure the peak-to-peak input voltage, v_{in1} , at TP50. You may use ACQUIRE, Average option of the oscilloscope to get a noise-free signal. Use the MEASURE, Pk-Pk option to read the value. Calculate the input resistance, R_{in} , of the amplifier assuming that the input impedance is purely real and using the resistive voltage divider formula:

$$v_{in1} = \frac{R_{in}}{R_{in} + 50} 0.100 \quad (8)$$

Note this value. You need this value in the preliminary part of Lab 6 experiment.

14. Measure the peak-to-peak output voltage, v_{o1} , at TP51, while the signal generator is still connected between TP50 and GND. Make sure that the output voltage is purely sinusoidal and that it is not distorted. Determine the voltage gain $|v_{o1}/v_{in1}|_{dB}$ in dB ($20\log_{10}()$). Compare it with the simulated result.

TP50, Measured $v_{in1}(pp) = 76.8 \mu V$ TP51, measured $v_{o1}(pp) = 4.36 V$
 $R_{in1} = 165.52 \Omega$ $|v_{o1}/v_{in1}|_{dB} = 35.62 dB$

Comparison: We measured more gain than we expected. Hence the discrepancy is not that large and could be due to measuring errors.

1.14. GRADE:

15. Connect the signal generator to TP60. Measure the peak-to-peak voltage at TP60 (v_{in2}) and TP61 (v_{o2}). Determine the voltage gain $|v_{o2}/v_{in2}|_{dB}$ in dB. Compare it with the simulated result.

TP60, Measured $v_{in2}(pp) = 74.2 \mu V$ TP61, measured $v_{o2}(pp) = 4.64 V$
 $R_{in2} = 145.13 \Omega$ $|v_{o2}/v_{in2}|_{dB} = 35.89 dB$

Comparison: Similar conclusions were reached to the previous entry. However, a slight increase in gain was observed, which matched the calculations.

1.15. GRADE:

16. Mount and solder C52 (10 nF) that connects the output of the first amplifier to the input of the second amplifier.

IF amplifier mounting is now finished. The testing of the combined IF amplifiers will be done later.

CHECK POINT: